



Customer Shopping Pattern Analysis

Uncovering insights from **3,900 transactions** across product categories, demographics, and subscription behavior.

Dataset at a Glance

3,900

Purchases

18

Columns

\$59.76

Avg. Purchase

3.75

Avg. Rating

Key features: demographics, purchase details, shipping, discounts, review ratings, and subscription status.

Data Preparation in Python

01

Load & Explore

Imported with `pandas`; used `df.info()` and `.describe()`

02

Clean & Standardize

Imputed 37 missing ratings; renamed columns to snake_case

03

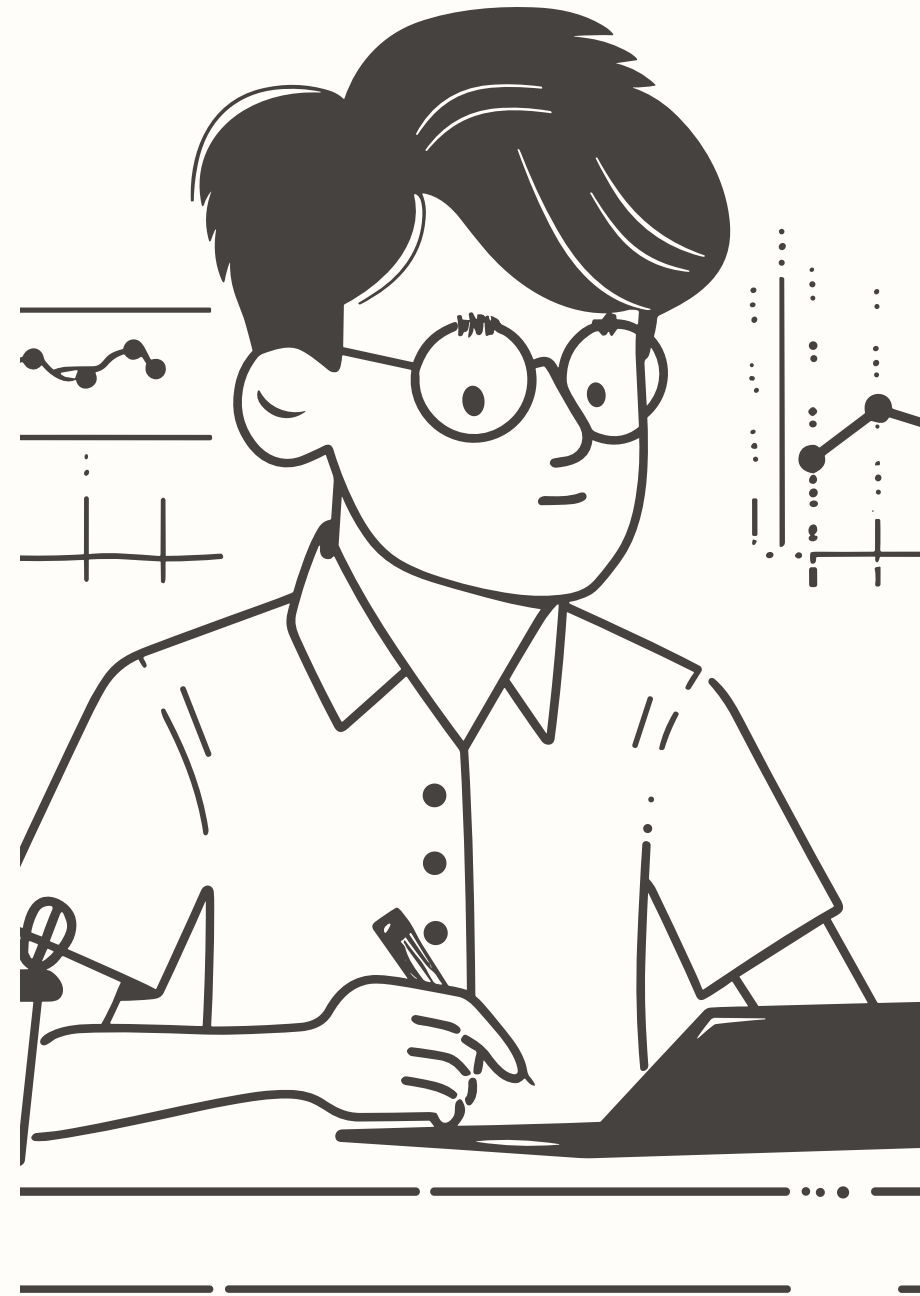
Feature Engineering

Created `age_group` and `purchase_frequency_days` columns

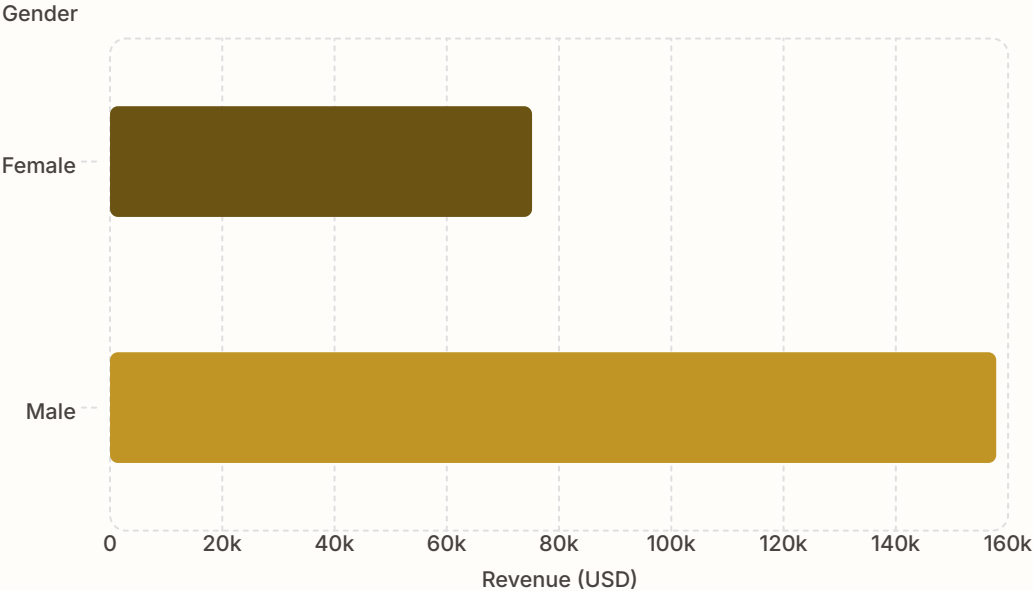
04

Load to MySQL

Connected Python to MySQL for SQL analysis



Revenue by Gender & Shipping



Shipping Type vs. Avg. Spend

Shipping	Avg. Amount
Standard	\$58.46
Express	\$60.48

Male customers generate **2× more revenue** than female.
Express shipping users spend slightly more.

Top-Rated Products & Discount Leaders

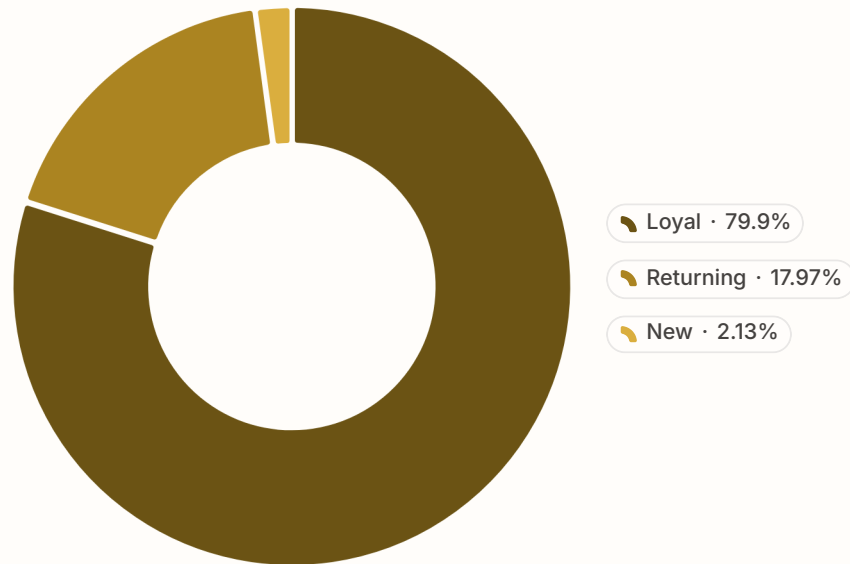
Top 5 by Avg. Rating

Product	Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discount-Dependent Products

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Customer Segmentation



Segment Breakdown

Loyal

3,116 customers —
dominant majority

Returning

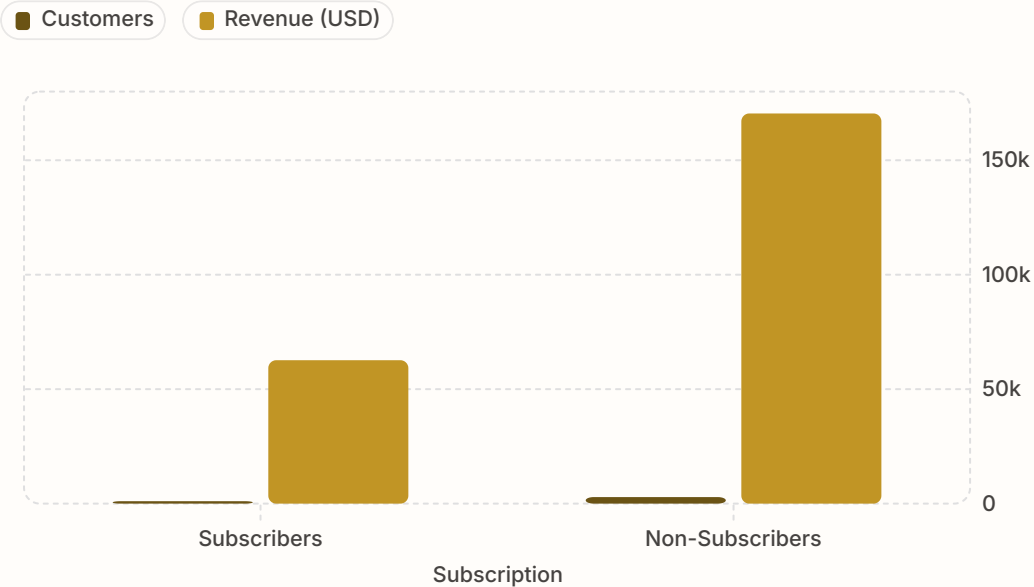
701 customers

New

83 customers

Customers with **>5 purchases**: 958 subscribed vs. 2,518 non-subscribed.

Subscribers vs. Non-Subscribers

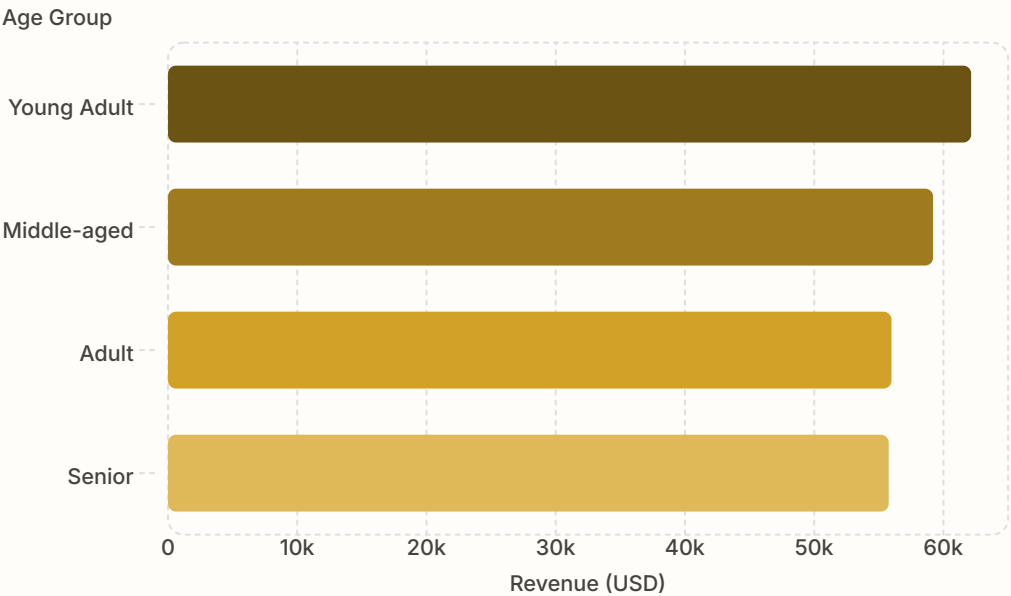


Avg. Spend Comparison

Status	Avg. Spend
Subscribers	\$59.49
Non-Subscribers	\$59.87

Spend per customer is nearly identical — **subscription drives volume, not ticket size.**

Revenue by Age Group & Top Products per Category



Top Products by Category

Category	#1 Product
Accessories	Jewelry (171)
Clothing	Blouse (171)
Footwear	Sandals (160)
Outerwear	Jacket (163)



Power BI Dashboard

Interactive dashboard consolidating all key metrics: subscription rates, revenue by category, age group performance, and sales trends.

Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers; move them to "Loyal" segment

Review Discount Policy

Balance sales boosts with margin control

Product Positioning

Highlight top-rated & best-selling items in campaigns

Targeted Marketing

Focus on Young Adults and express-shipping users